

Patient



Sex: Male

Case Number: TN24-206675

Diagnosis: Non-small cell carcinoma

Specimen Information

Primary Tumor Site: Upper lobe, lung

Specimen Site: Upper lobe, lung

Specimen ID: C24-09362-C1

Specimen Collected: 17-Jul-2024

Test Report Date: 09-Aug-2024

Ordered By

Ashish Maskey

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Results with Therapy Associations

BIOMARKER	METHOD	ANALYTE	RESULT	THERAPY ASSOCIATION		BIOMARKER LEVEL*
PD-L1 (22c3)	IHC	Protein	Positive, TPS: 90%	BENEFIT	cemiplimab, pembrolizumab	Level 1
PD-L1 (28-8)	IHC	Protein	Positive 2+, 90%	BENEFIT	nivolumab/ipilimumab combination	Level 1
PD-L1 (SP263)	IHC	Protein	Positive, TC: 2+, 90%	BENEFIT	atezolizumab (adjuvant), cemiplimab	Level 1
EGFR	Seq	DNA-Tumor	Pathogenic Variant Exon 18 p.G719A	BENEFIT	afatinib	Level 2
					osimertinib	Level 2
					dacomitinib, erlotinib, gefitinib	Level 3
TMB	Seq	DNA-Tumor	High, 25 mut/Mb	BENEFIT	pembrolizumab	Level 2
ALK	Seq	RNA-Tumor	Fusion Not Detected	LACK OF BENEFIT	alectinib, brigatinib, ceritinib, crizotinib, lorlatinib	Level 2
BRAF	Seq	DNA-Tumor	Mutation Not Detected	LACK OF BENEFIT	dabrafenib, dabrafenib + trametinib, encorafenib + binimetinib, vemurafenib	Level 2
KRAS	Seq	DNA-Tumor	Mutation Not Detected	LACK OF BENEFIT	adagrasib, sotorasib	Level 2
MET	Seq	RNA-Tumor	Variant Transcript Not Detected	LACK OF BENEFIT	capmatinib, tepotinib	Level 2
					crizotinib	Level 2
RET	Seq	RNA-Tumor	Fusion Not Detected	LACK OF BENEFIT	pralsetinib, selpercatinib	Level 2
ROS1	Seq	RNA-Tumor	Fusion Not Detected	LACK OF BENEFIT	ceritinib, crizotinib, entrectinib, lorlatinib, repotrectinib	Level 2

* Biomarker reporting classification: Level 1 – Companion diagnostic (CDx); Level 2 – Strong evidence of clinical significance or is endorsed by standard clinical guidelines; Level 3 – Potential clinical significance. Bolded benefit therapies, if present, highlight the most clinically significant findings.

The selection of any, all, or none of the matched therapies resides solely with the discretion of the treating physician. Decisions on patient care and treatment must be based on the independent medical judgment of the treating physician, taking into consideration all available information concerning the patient's condition, the FDA prescribing information for any therapeutic, and in accordance with the applicable standard of care. Whether or not a particular patient will benefit from a selected therapy is based on many factors and can vary significantly. All trademarks and registered trademarks are the property of their respective owners.

Important Note

This patient's tumor is positive for an activating EGFR mutation, PD-L1 expression and is TMB-High. The use of EGFR-targeted tyrosine kinase inhibitors (TKIs) should be prioritized over immune checkpoint inhibitors (ICIs) for patients with activating mutations in EGFR. Immune checkpoint therapy could be considered after EGFR-targeted treatment options have been exhausted. Please see NCCN guidelines and clinicaltrials.gov for more information.

Please note that multiple companion diagnostic assays (antibodies) have been utilized to assess PD-L1 expression. Each test has different performance characteristics, therefore, the results will not always be concordant.

For information on Caris GPSai[™], please refer to relevant section of the report (Additional Results Page 1).

Cancer-Type Relevant Biomarkers

Biomarker	Method	Analyte	Result
TP53	Seq	DNA-Tumor	Pathogenic Variant Exon 7 p.S241T
		DNA-Tumor	Pathogenic Variant Exon 7 p.S241F
MSI	Seq	DNA-Tumor	Stable
NTRK1/2/3	Seq	RNA-Tumor	Fusion Not Detected
ALK	Seq	DNA-Tumor	Mutation Not Detected
BRAF	Seq	RNA-Tumor	Fusion Not Detected
EGFR	Seq	DNA-Tumor	Variant of Uncertain Significance Exon 21 p.G824A
ERBB2 (Her2/Neu)	CNA-Seq	DNA-Tumor	Amplification Not Detected
	IHC	Protein	Negative 0
	Seq	DNA-Tumor	Mutation Not Detected

Biomarker	Method	Analyte	Result
FGFR3	Seq	RNA-Tumor	Fusion Not Detected
KEAP1	Seq	DNA-Tumor	Mutation Not Detected
MET	CNA-Seq	DNA-Tumor	Amplification Not Detected
	Seq	DNA-Tumor	Mutation Not Detected
MTAP	CNA-Seq	DNA-Tumor	Deletion Not Detected
NFE2L2	Seq	DNA-Tumor	Mutation Not Detected
NRG1	Seq	RNA-Tumor	Fusion Not Detected
PD-L1 (SP142)	IHC	Protein	Negative, IC: 0% Negative, TC: 2+, 20%
RB1	Seq	DNA-Tumor	Indeterminate
RET	Seq	DNA-Tumor	Mutation Not Detected
STK11	Seq	DNA-Tumor	Mutation Not Detected

Genomic Signatures

Biomarker	Method	Analyte	Result
Microsatellite Instability (MSI)	Seq	DNA-Tumor	Stable
Tumor Mutational Burden (TMB)	Seq	DNA-Tumor	Result: High 25 Low 10 High
Genomic Loss of Heterozygosity (LOH)	Seq	DNA-Tumor	Equivocal - 14% of tested genomic segments exhibited LOH (assay threshold is ≥ 16%)

PATIENT: [REDACTED]

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Genes Tested with Pathogenic or Likely Pathogenic Alterations

Gene	Method	Analyte	Variant Interpretation	Protein Alteration	Exon	DNA Alteration	Variant Frequency %
EGFR	Seq	DNA-Tumor	Pathogenic Variant	p.G719A	18	c.2156G>C	42
PTEN	Seq	DNA-Tumor	Pathogenic Variant	p.Q110*	5	c.328C>T	29
RBM10	Seq	DNA-Tumor	Pathogenic Variant	p.E47*	3	c.139G>T	35
TERT	Seq	DNA-Tumor	Pathogenic Variant	-	0	c.-124C>T	32
TP53	Seq	DNA-Tumor	Pathogenic Variant	p.S241T	7	c.721T>A	12
	Seq	DNA-Tumor	Pathogenic Variant	p.S241F	7	c.722C>T	24

Unclassified alterations for DNA and RNA sequencing can be found in the MI Portal.

Formal nucleotide nomenclature and gene reference sequences can be found in the Appendix of this report.

Genes Tested with Variants of Uncertain Significance

Gene	Method	Analyte	Variant Interpretation	Protein Alteration	Exon	DNA Alteration	Variant Frequency %
BRCA2	Seq	DNA-Tumor	Variant of Uncertain Significance	p.S662F	11	c.1985C>T	19
	Seq	DNA-Tumor	Variant of Uncertain Significance	p.N570D	10	c.1708A>G	13
EGFR	Seq	DNA-Tumor	Variant of Uncertain Significance	p.G824A	21	c.2471G>C	40
PDGFRA	Seq	DNA-Tumor	Variant of Uncertain Significance	p.K226T	5	c.677A>C	24
RBM10	Seq	DNA-Tumor	Variant of Uncertain Significance	p.I334M	10	c.1002A>G	40

Additional Variants of Uncertain Significance can be found in the MI Portal.

Human Leukocyte Antigen (HLA) Genotype Results

The impact of HLA genotypes on drug response and prognosis is an active area of research. These results can help direct patients to clinical trials recruiting for specific genotypes. Please see www.clinicaltrials.gov for more information.

Gene	Method	Analyte	Genotype
MHC CLASS I			
HLA-A	Seq	DNA-Tumor	A*03:01, A*11:01
HLA-B	Seq	DNA-Tumor	B*40:01, B*44:03
HLA-C	Seq	DNA-Tumor	C*16:01, C*03:14

HLA genotypes with only one allele are either homozygous or have loss-of-heterozygosity at that position.

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Immunohistochemistry Results

Biomarker	Result	Biomarker	Result
ERBB2 (Her2/Neu)	Negative 0	PD-L1 (SP142)	Negative, IC: 0% Negative, TC: 2+, 20%
PD-L1 (22c3)	Positive, TPS: 90%	PD-L1 (SP263)	Positive, TC: 2+, 90%
PD-L1 (28-8)	Positive 2+, 90%		

Genes Tested with Indeterminate Results by Tumor DNA Sequencing

ATP6AP2	COL2A1	EED	JAK2	MDM2	PLCB4	PRKD1	RASA1	RPA2	SMARCE1	STAG2	TRRAP
ATRX	CUL3	EGLN1	KIF1B	MGA	PMS1	PTPRD	RB1	RRAS2	SOS1	SUZ12	WRN
B2M	DACH1	EIF1AX	KMT2C	NOTCH2	POLQ	RABL3	REST	SMARCA2	SSBP1	TRIM28	XRCC2
CDK6	DGCR8	EXO1	MDH2	NPM1	PRDM6	RAD50					

Genes in this table were ruled indeterminate due to low coverage for some or all exons.

Genes Tested with Intermediate CNA Results by Tumor DNA Sequencing

CCND1	FGF3	FGF4									
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The results in this report were curated to represent biomarkers most relevant for the submitted cancer type. These include results important for therapeutic decision-making, as well as notable alterations in other biomarkers known to be involved in oncogenesis. Additional results, including genes with normal findings, variants of uncertain significance, or unclassified alterations can be found in the MI Portal at miportal.carismolecularintelligence.com. If you do not have an MI Portal account, or need assistance accessing it, please contact Caris Customer Support at (888) 979-8669.

Notes of Significance

SEE APPENDIX FOR DETAILS

Clinical Trials Connector™ opportunities based on biomarker expression: 500 Targeted Therapy Trials. See page 6 for details.

Specimen Information

Specimen ID: C24-09362-C1

Specimen Collected: 07/17/2024

Specimen Received: 07/27/2024

Testing Initiated: 07/30/2024

Test Ordered*: MI Profile™ (MI Tumor Seek Hybrid™ + IHCs and Other Tests by Tumor Type)

* If the submitted specimen is inadequate, only a subset of the ordered testing may be reported.

Gross Description: 1 (A) Paraffin Block - Client ID(C24-09362-C1) from UK HealthCare - Albert B Chandler Hospital, Lexington, KY, with the corresponding pathology report labeled "C24-09362".

Dissection Information: Molecular testing of this specimen was performed after harvesting of targeted tissues with an approved manual microdissection technique. Candidate slides were examined under a microscope and areas containing tumor cells (and separately normal cells, when necessary for testing) were circled. A laboratory technician harvested targeted tissues for extraction from the marked areas using a dissection microscope.

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Clinical Trials Connector™

For a complete list of open, enrolling clinical trials visit MI Portal to access the [Clinical Trials Connector](#). This personalized, real-time web-based service provides additional clinical trial information and enhanced searching capabilities, including, but not limited to:

- Location: filter by geographic area
- Biomarker(s): identify specific biomarkers associated with open clinical trials to choose from
- Drug(s): search for specific therapies
- Trial Sponsor: locate trials based on the organization supporting the trial(s)

The Clinical Trials Connector lists agents that are matched to available clinical trials according to biomarker status. In some instances, older-generation agents may still be relevant in the context of new combination strategies and, therefore, will still appear on this report.

Visit www.Carismolecularintelligence.com to view all matched trials. Therapeutic agents listed below may or may not be currently FDA approved for the tumor type tested.

TARGETED THERAPY CLINICAL TRIALS (500)				
Drug Class	Biomarker	Method	Analyte	Investigational Agent(s)
EGFR TKIs (59)	EGFR	NGS	DNA-Tumor	CLN-081, WSD0922-FU, erlotinib, gefitinib, lapatinib, lazertinib, nazartinib, osimertinib
Immunomodulatory agents (388)	TMB	NGS	DNA-Tumor	INBRX-105, M7824, MGD019, atezolizumab, avelumab, camrelizumab, cemiplimab, cetrelimab, dostarlimab, durvalumab, ipilimumab, nivolumab, pembrolizumab, sintilimab, spartalizumab, tislelizumab, toripalimab, tremelimumab
	PD-L1	IHC	Protein	
PARP inhibitors (32)	PTEN	NGS	DNA-Tumor	2X-121, BGB-290, niraparib, olaparib, pamiparib, talazoparib
PI3K/Akt/mTOR inhibitors (21)	PTEN	NGS	DNA-Tumor	CYH33, afuresertib, alpelisib, buparlisib, capivasertib, everolimus, gedatolisib, inavolisib, ipatasertib, nab-rapamycin, nab-sirolimus, sirolimus, temsirolimus

() = represents the total number of clinical trials identified by the Clinical Trials Connector for the provided drug class or table.

The Clinical Trials Connector may include trials that enroll patients with additional screening of molecular alterations. In some instances, only specific gene variants may be eligible.

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Disclaimer

Decisions on patient care and treatment must be based on the independent medical judgment of the treating physician, taking into consideration all available information concerning the patient's condition, prescribing information for any therapeutic, and in accordance with the applicable standard of care. Drug associations provided in this report do not guarantee that any particular agent will be effective for the treatment of any patient or for any particular condition. Caris Life Sciences® expressly disclaims and makes no representation or warranty whatsoever relating, directly or indirectly, to the performance of services, including any information provided and/or conclusions drawn from therapies that are included or omitted from this report. Whether or not a particular patient will benefit from a selected therapy is based on many factors and can vary significantly. The selection of therapy, if any, resides solely in the discretion of the treating physician and the tests should not be considered a companion diagnostic.

Caris MPI, Inc. d/b/a Caris Life Sciences is certified under the Clinical Laboratory Improvement Amendments (CLIA) as qualified to perform high complexity clinical laboratory testing, including all Caris molecular profiling assays. Individual assays that are available through Caris molecular profiling include both Laboratory Developed Tests (LDT) and U.S. Food and Drug Administration (FDA) approved or cleared tests. In addition, certain tests have been CE-marked as a general IVD under the In Vitro Diagnostic Directive (IVDD) 98/79/EC. Offered LDTs were developed and their performance characteristics determined by Caris. Certain tests have not been cleared or approved by the FDA. Caris LDTs are used for clinical purposes. They are not investigational or for research. Caris' CLIA certification number is located at the bottom of each page of this report.

The information presented in the Clinical Trials Connector™ section of this report, if applicable, is compiled from sources believed to be reliable and current. However, the accuracy and completeness of the information provided herein cannot be guaranteed. The clinical trials information present in the biomarker description was compiled from www.clinicaltrials.gov. The contents are to be used only as a guide, and health care providers should employ their best comprehensive judgment in interpreting this information for a particular patient. Specific eligibility criteria for each clinical trial should be reviewed as additional inclusion criteria may apply.

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Caris molecular testing is subject to Caris' intellectual property. Patent www.CarisLifeSciences.com/ip.

Professional Component Performed: -- Location 1, Chandler, AZ. Jose de los Santos, MD, Lab Director.

A handwritten-style electronic signature in black ink, reading "Jose de los Santos, M.D.".

Electronic Signature

Jose de los Santos, MD

09-Aug-2024



The Caris GPSai™ (Caris Genomic Probability Score – Artificial Intelligence) is a cancer-type similarity assessment which compares the characteristics of a patient's tumor against other tumors in the Caris database. Caris GPSai analyzes a tumor's molecular signature and provides the prevalence of that signature in the Caris Life Sciences genomic and transcriptomic database across 90 distinct cancer categories.

Most Likely Tumor Type: Non-Small Cell Lung Carcinoma

94%

The predicted probability of each cancer type that can be detected by the product are shown below. The total probabilities of the top level cancer type (blue) total 100%. The subtypes (indented) sum to the total probability predicted for the top level cancer type. Discrepancies between expected sums and displayed sums may be exhibited due to rounding.

0	Adrenal Cortical Carcinoma
2	Bladder/Urinary Tract
2	Urothelial Carcinoma
0	Bladder Adenocarcinoma
0	Bowel
0	Appendiceal Adenocarcinoma
0	Colorectal Adenocarcinoma
0	Small Bowel Carcinoma
0	Breast
0	Metaplastic Breast Cancer
0	Breast Invasive Lobular Carcinoma
0	Breast Invasive Ductal Carcinoma
0	CNS/Brain
0	Diffuse Glioma
0	Meningioma
0	Cervix/Uterine Carcinoma
0	Uterine Serous Carcinoma/Uterine Papillary Serous Carcinoma
0	Uterine Carcinosarcoma/Uterine Malignant Mixed Mullerian Tumor
0	Uterine Clear Cell Carcinoma
0	Cervical Adenocarcinoma
0	Uterine Endometrioid Carcinoma
1	Cutaneous Squamous Cell Carcinoma
1	Esophagus/Stomach
0	Esophageal Squamous Cell Carcinoma
0	Esophagogastric Adenocarcinoma
0	Stomach Adenocarcinoma
0	Adenocarcinoma of the Gastroesophageal Junction
0	Esophageal Adenocarcinoma
0	Germ Cell Tumor
0	Hematological
0	Hepatocellular Carcinoma
0	Kidney
0	Renal Cell Carcinoma
0	Chromophobe Renal Cell Carcinoma
0	Papillary Renal Cell Carcinoma
0	Renal Clear Cell Carcinoma
0	Wilms Tumor
0	Melanoma
0	Mesothelioma
0	Neuroendocrine Neoplasm
0	Well/Moderately-Differentiated Neuroendocrine Tumor
0	Poorly-Differentiated Neuroendocrine Carcinoma
0	Large Cell Neuroendocrine Carcinoma
0	Small Cell Neuroendocrine Carcinoma
0	Paraganglioma/Pheochromocytoma
0	Merkel Cell Carcinoma

94	Non-Small Cell Lung Carcinoma
89	Lung Squamous Cell Carcinoma
6	Lung Adenocarcinoma
2	Orogenital Squamous Cell Carcinoma
0	Ovarian Epithelial Tumor
0	Ovarian Carcinosarcoma/Malignant Mixed Mesodermal Tumor
0	Serous Ovarian/Fallopian Tube/Peritoneal
0	Low-Grade Serous Ovarian/Fallopian Tube/Peritoneal Cancer
0	High-Grade Serous Ovarian/Fallopian Tube/Peritoneal Cancer
0	Mucinous Ovarian Cancer
0	Clear Cell Ovarian Cancer
0	Endometrioid Ovarian Cancer
0	Pancreatobiliary
0	Gallbladder Cancer
0	Pancreatic Adenocarcinoma
0	Cholangiocarcinoma
0	Peripheral Nervous System
0	Schwannoma
0	Neuroblastoma
0	Malignant Peripheral Nerve Sheath Tumor
0	Prostate Adenocarcinoma
0	Salivary Gland Tumor
0	Sex Cord Stromal Tumor
0	Granulosa Cell Tumor
0	Sertoli-Leydig Cell Tumor
0	Soft Tissue/Bone
0	Gastrointestinal Stromal Tumor
0	Leiomyosarcoma
0	Angiosarcoma
0	Rhabdomyosarcoma
0	Synovial Sarcoma
0	Osteosarcoma
0	Liposarcoma
0	Chondrosarcoma
0	Endometrial Stromal Sarcoma
0	Ewing Sarcoma
0	Thymic Carcinoma
0	Thyroid
0	Medullary Thyroid Cancer
0	Hurthle Cell Thyroid Cancer
0	Well-Differentiated Thyroid Cancer
0	Follicular Thyroid Cancer
0	Papillary Thyroid Cancer
0	Anaplastic Thyroid Cancer

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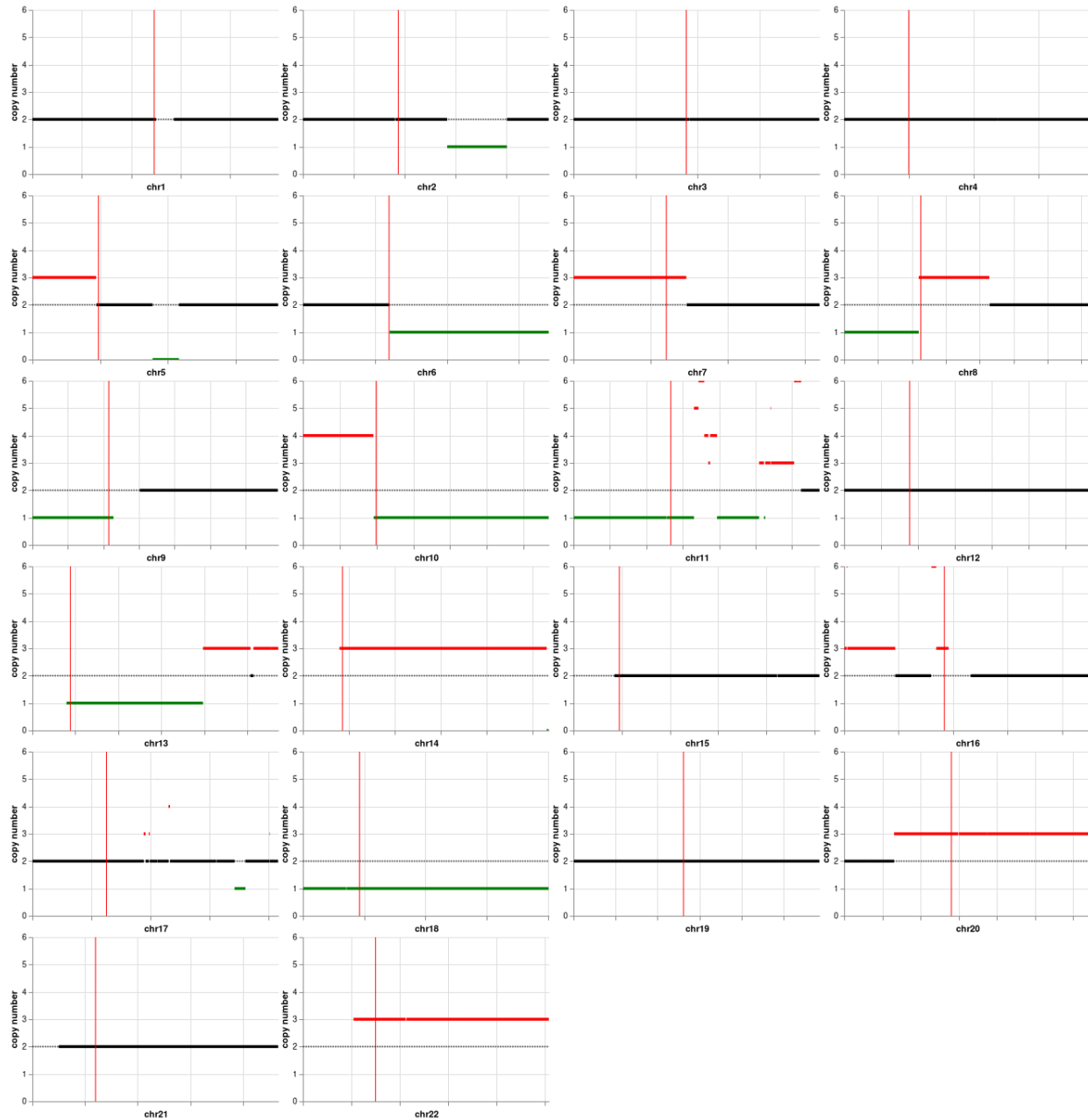
Gene Expression

Gene	Percentile in Cancer Type	Gene	Percentile in Cancer Type	Gene	Percentile in Cancer Type
ADORA2A	62	FOLR1	73	NTRK3	57
ALK	14	GPC3	6	PDCD1	63
ASCL1	58	HGF	20	PDCD1LG2	77
ATM	74	HRAS	66	PIK3CA	4
AURKA	90	IGF1R	64	POU3F2	50
BRAF	18	KDM1A	29	PRAME	22
BRCA1	72	KDR	29	PTEN	32
BRCA2	66	KEAP1	48	RB1	44
BRD4	84	KRAS	34	RET	17
CCND1	94	LAG3	64	ROR1	76
CCND2	35	MAGEA4	96	ROR2	98
CCNE1	72	MDM2	10	ROS1	50
CD274	90	MET	48	SLFN11	42
CD276	100	MKI67	82	SRC	24
CDKN2A	10	MSLN	35	SSTR2	28
CEACAM5	6	MTAP	90	SSTR3	30
CLDN18	19	MTOR	39	SSTR5	85
CLDN6	41	MUC1	55	STK11	52
CTLA4	47	MUC16	50	TACSTD2	86
DLL3	20	MYC	87	TGFB1	42
EGFR	88	NECTIN4	40	TOP1	36
EPHA2	57	NEUROD1	53	TP53	84
ERBB2	48	NF1	28	TSC1	41
ERBB3	24	NRAS	54	TSC2	64
FGFR1	24	NRG1	6	VEGFA	83
FGFR2	6	NTRK1	50	XPO1	23
FGFR3	24	NTRK2	20	YAP1	96

Gene Expression of Selected Genes by Whole Transcriptome Sequencing (WTS) Methods:

Gene expression is derived from whole transcriptome sequencing. Relative expression of genes are calculated as normalized values using Transcripts per Million Molecules or TPM. TPM is presented as a percentile derived by comparison to a distribution of Caris' internal cohort of the tumor-type profiled. Selected genes reported in this section were chosen based on their tumor-type specific relevance for matching to clinical trials, or tumor type subclassification.

Karyotype



Karyotyping using Copy Number Analysis by Whole Exome Sequencing (WES) Methods:

Whole exome sequencing in combination with interrogation of single nucleotide polymorphisms (SNPs) tiled throughout the genome, allows for the identification and visualization of cytogenetic aberrations.

Somatic structural variants like whole or partial chromosome duplications or deletions, are important for cancer development and progression, and may identify clinically actionable alterations.

Mutational Analysis by Next-Generation Sequencing (NGS)

TUMOR MUTATIONAL BURDEN	
Mutations / Megabase	Result
25	High

TMB

Tumor Mutational Burden (TMB) is defined as the number of somatic non-synonymous mutations per million bases of sequenced DNA in a tumor sample. Tumors with high TMB may increase the number of neoantigens which is hypothesized to increase T-cell reactivity and potential for response to immune checkpoint inhibitors. TMB analysis was performed based on next generation sequencing analysis of genomic DNA isolated from a tumor sample.

MICROSATELLITE INSTABILITY ANALYSIS	
Test	Result
MSI	Stable

MSI

Microsatellite instability (MSI) status is a measure of the number of somatic mutations within short, repeated sequences of DNA (microsatellites). MSI-High status can indicate that the tumor has a defect in mismatch repair (MMR) abrogating the ability to correct mistakes during DNA replication. Tumors with MSI-high status may increase the number of neoantigens which is hypothesized to increase T-cell reactivity and potential for response to immune checkpoint inhibitors. Tumor-only microsatellite instability status by NGS (MSI-NGS) is measured by the direct analysis of known microsatellite regions sequenced in the CMI NGS panel.

GENOMIC LOSS OF HETEROZYGOSITY	
Test	Result
Genomic Loss of Heterozygosity (LOH)	Equivocal - 14% of tested genomic segments exhibited LOH (assay threshold is $\geq 16\%$)

LOH

To calculate genomic loss-of-heterozygosity (LOH), the 22 autosomal chromosomes are split into 552 segments and the LOH of single nucleotide polymorphisms (SNPs) within each segment is calculated. Caris WES data consist of approximately 250k SNPs spread across the genome. SNP alleles with frequencies skewed towards 0 or 100% indicate LOH (heterozygous SNP alleles have a frequency of 50%). The final call of genomic LOH is based on the percentage of all 552 segments with observed LOH.

Additional Next-Generation Sequencing results continued on the next page. >

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Mutational Analysis by Next-Generation Sequencing (NGS)

Gene	Analyte	Variant Interpretation	Protein Alteration	Exon	DNA Alteration	Variant Frequency %	Transcript ID
BRCA2	DNA-Tumor	Variant of Uncertain Significance	p.S662F	11	c.1985C>T	19	NM_000059.3

Interpretation: This variant has not been reported in the literature. As such, its clinical significance is not currently known.

BRCA2 or breast cancer type 2 susceptibility gene encodes a protein involved in cell growth, cell division, and DNA-damage repair. It is a tumor suppressor gene which plays an important role in mediating double-strand DNA breaks by homologous recombination (HR). Tumors with BRCA2 mutation may be more sensitive to platinum agents and PARP inhibitors.

Gene	Analyte	Variant Interpretation	Protein Alteration	Exon	DNA Alteration	Variant Frequency %	Transcript ID
BRCA2	DNA-Tumor	Variant of Uncertain Significance	p.N570D	10	c.1708A>G	13	NM_000059.3

Interpretation: A variant with no known clinical or functional significance was detected in BRCA2.

Gene	Analyte	Variant Interpretation	Protein Alteration	Exon	DNA Alteration	Variant Frequency %	Transcript ID
EGFR	DNA-Tumor	Variant of Uncertain Significance	p.G824A	21	c.2471G>C	40	NM_005228.4

Interpretation: A variant with no known clinical or functional significance was detected in EGFR.

EGFR or epidermal growth factor receptor, is a transmembrane receptor tyrosine kinase belonging to the ErbB family of receptors. Upon ligand binding, the activated receptor triggers a series of intracellular pathways (Ras/MAPK, PI3K/Akt, JAK-STAT) that result in cell proliferation, migration and adhesion. EGFR mutations have been observed in 20-25% of non-small cell lung cancer (NSCLC), 10% of endometrial and peritoneal cancers. Somatic gain-of-function EGFR mutations, including in-frame deletions in exon 19 or point mutations in exon 21, confer sensitivity to first- and second-generation tyrosine kinase inhibitors (TKIs), whereas the secondary mutation, T790M in exon 20, confers reduced response. Germline mutations and polymorphisms of EGFR have been associated with familial lung adenocarcinomas.

Gene	Analyte	Variant Interpretation	Protein Alteration	Exon	DNA Alteration	Variant Frequency %	Transcript ID
EGFR	DNA-Tumor	Pathogenic Variant	p.G719A	18	c.2156G>C	42	NM_005228.4

Interpretation: A pathogenic G719 mutation was detected in EGFR exon 18.

Gene	Analyte	Variant Interpretation	Protein Alteration	Exon	DNA Alteration	Variant Frequency %	Transcript ID
PDGFRA	DNA-Tumor	Variant of Uncertain Significance	p.K226T	5	c.677A>C	24	NM_006206.5

Interpretation: This is a rare variant with unknown clinical or biological significance.

PDGFRA is the alpha-type platelet-derived growth factor receptor, a surface tyrosine kinase receptor structurally homologous to c-KIT, which activates PI3KCA/AKT, RAS/MAPK and JAK/STAT signaling pathways. PDGFRA mutations are found in 5-8% of patients with gastrointestinal stromal tumors (GIST) and increases to 30% in KIT wildtype GIST. Germline mutations in PDGFRA have been associated with Familial gastrointestinal stromal tumors and Hypereosinophilic Syndrome (HES).

Additional Next-Generation Sequencing results continued on the next page. >

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Mutational Analysis by Next-Generation Sequencing (NGS)

Gene	Analyte	Variant Interpretation	Protein Alteration	Exon	DNA Alteration	Variant Frequency %	Transcript ID
PTEN	DNA-Tumor	Pathogenic Variant	p.Q110*	5	c.328C>T	29	NM_000314.6

Interpretation: A pathogenic mutation was detected in PTEN. Germline mutations in the PTEN gene are causal for PTEN Hamartoma Tumor Syndrome (Cowden syndrome).

PTEN or phosphatase and tensin homolog is a tumor suppressor gene that prevents cells from proliferating. PTEN is an important mediator in signaling downstream of EGFR, and loss of PTEN gene function/expression due to gene mutations or allele loss is associated with reduced benefit to EGFR-targeted monoclonal antibodies. Mutation in PTEN is found in 5-14% of colorectal cancer and 7% of breast cancer. PTEN mutation leads to loss of function of the encoded phosphatase, and an upregulation of the PIK3CA/AKT pathway. Germline PTEN mutations associate with Cowden disease and Bannayan-Riley-Ruvalcaba syndrome. These dominantly inherited disorders belong to a family of hamartomatous polyposis syndromes which feature multiple tumor-like growths (hamartomas) accompanied by an increased risk of breast carcinoma, follicular carcinoma of the thyroid, glioma, prostate and endometrial cancer. Trichilemmoma, a benign, multifocal neoplasm of the skin is also associated with PTEN germline mutations.

Gene	Analyte	Variant Interpretation	Protein Alteration	Exon	DNA Alteration	Variant Frequency %	Transcript ID
RBM10	DNA-Tumor	Variant of Uncertain Significance	p.I334M	10	c.1002A>G	40	NM_005676.4

Interpretation: A variant with no known clinical or functional significance was detected in RBM10.

Gene	Analyte	Variant Interpretation	Protein Alteration	Exon	DNA Alteration	Variant Frequency %	Transcript ID
RBM10	DNA-Tumor	Pathogenic Variant	p.E47*	3	c.139G>T	35	NM_005676.4

Interpretation: A pathogenic variant was detected in RBM10.

Gene	Analyte	Variant Interpretation	Protein Alteration	Exon	DNA Alteration	Variant Frequency %	Transcript ID
TERT	DNA-Tumor	Pathogenic Variant	-	0	c.-124C>T	32	NM_198253.2

Interpretation: A pathogenic variant was detected in TERT promoter region, also referred to as C228T in hg19.

Telomerase is a ribonucleoprotein polymerase that maintains telomere ends by addition of the telomere repeat TTAGGG. Telomerase expression plays a role in cellular senescence, as it is normally repressed in postnatal somatic cells resulting in progressive shortening of telomeres. Deregulation of telomerase expression in somatic cells may be involved in oncogenesis. Studies in mouse suggest that telomerase also participates in chromosomal repair, since de novo synthesis of telomere repeats may occur at double-stranded breaks.

Additional Next-Generation Sequencing results continued on the next page. >

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TN24-206675

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Mutational Analysis by Next-Generation Sequencing (NGS)

Gene	Analyte	Variant Interpretation	Protein Alteration	Exon	DNA Alteration	Variant Frequency %	Transcript ID
TP53	DNA-Tumor	Pathogenic Variant	p.S241T	7	c.721T>A	12	NM_000546.5

Interpretation: A TP53 mutation, S241T, was detected. This mutation, and other substitutions at the same position in the protein, have been identified in numerous tumors. S241T has also been reported as a germline mutation, causal for Li-Fraumeni syndrome (Monti 2011 Mol Cancer Res 9:271).

TP53, or p53, plays a central role in modulating response to cellular stress through transcriptional regulation of genes involved in cell-cycle arrest, DNA repair, apoptosis, and senescence. Inactivation of the p53 pathway is essential for the formation of the majority of human tumors. Mutation in p53 (TP53) remains one of the most commonly described genetic events in human neoplasia, estimated to occur in 30-50% of all cancers. Generally, presence of a disruptive p53 mutation is associated with a poor prognosis in all types of cancers, and diminished sensitivity to radiation and chemotherapy. Germline p53 mutations are associated with the Li-Fraumeni syndrome (LFS) which may lead to early-onset of several forms of cancer currently known to occur in the syndrome, including sarcomas of the bone and soft tissues, carcinomas of the breast and adrenal cortex (hereditary adrenocortical carcinoma), brain tumors and acute leukemias.

Gene	Analyte	Variant Interpretation	Protein Alteration	Exon	DNA Alteration	Variant Frequency %	Transcript ID
TP53	DNA-Tumor	Pathogenic Variant	p.S241F	7	c.722C>T	24	NM_000546.5

Interpretation: A pathogenic mutation, p.S241F, was detected in TP53. Substitutions at codon 241 are common in cancers. p.S241F has been reported as a germline mutation, causal for Li-Fraumeni syndrome (Monti 2011 Mol Cancer Res 9:271).

Additional Next-Generation Sequencing results continued on the next page. >

TN24-206675

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Mutational Analysis by Next-Generation Sequencing (NGS)

GENES TESTED WITH INDETERMINATE* RESULTS BY TUMOR DNA SEQUENCING

ATP6AP2	EED	MDM2	PRKD1	RRAS2	TRRAP
ATRX	EGLN1	MGA	PTPRD	SMARCA2	WRN
B2M	EIF1AX	NOTCH2	RABL3	SMARCE1	XRCC2
CDK6	EXO1	NPM1	RAD50	SOS1	
COL2A1	JAK2	PLCB4	RASA1	SSBP1	
CUL3	KIF1B	PMS1	RB1	STAG2	
DACH1	KMT2C	POLQ	REST	SUZ12	
DGCR8	MDH2	PRDM6	RPA2	TRIM28	

* Genes in this table were ruled indeterminate due to low coverage for some or all exons.

For a complete list of genes tested, visit www.CarisMolecularIntelligence.com/profilemenu.

NGS Methods

Direct sequence analysis was performed on genomic DNA isolated from a micro-dissected tumor sample using Illumina NovaSeq 6000 sequencers. A hybrid pull-down panel of baits was used to enrich more than 700 clinically relevant genes along with > 20,000 other genes. Sequence data is analyzed using a customized bioinformatics pipeline to detect sequencing variants, copy number alterations (amplifications and deletions) indels and HLA genotypes. In addition, genomic signatures for tumor mutational burden (TMB), microsatellite instability (MSI), genomic loss-of-heterozygosity (LOH) or HRD-Genomic Scar Score (HRD-GSS), and homologous recombination deficiency (HRD) are reported when applicable. For a complete list of what is covered by the assay, and genes with partial coverage, please contact Caris Customer Support.

Copy Number Alterations by Next-Generation Sequencing (NGS)

GENES TESTED WITH INTERMEDIATE CNA RESULTS					
CCND1	FGF3	FGF4			

CNA Methods

The copy number alteration (CNA) of each exon is determined by a calculation using the average sequencing depth of the sample along with the sequencing depth of each exon and comparing this calculated result to a pre-calibrated value. A complete list of genes for reporting copy number alterations, including amplifications and deletions, is available upon request.

Gene Fusion and Transcript Variant Detection by RNA Sequencing

Whole Transcriptome Sequencing (WTS) Methods

Gene fusion and variant transcript detection were performed on RNA isolated from a tumor sample using next generation sequencing. The assay also detects fusions occurring at known and novel breakpoints within genes. The genes included in this report represent the subset of genes associated with cancer. The complete list of unclassified alterations is available by request.

Protein Expression by Immunohistochemistry (IHC)

PD-L1 TUMOR CELL STAINING				
Biomarker	Patient Tumor			Thresholds
	Staining Intensity (0, 1+, 2+, 3+)	Percent of cells	Result	Conditions for a Positive Result:
PD-L1 (28-8)	2 +	90%	Positive	Intensity $\geq 1+$ and $\geq 1\%$ of cells stained
PD-L1 (SP142)	2 +	20%	Negative	Intensity $\geq 1+$ and $\geq 50\%$ of cells stained
PD-L1 (SP263)	2 +	90%	Positive	Intensity $\geq 1+$ and $\geq 1\%$ of cells stained

PD-L1 (28-8): Scoring was based on percentage of viable tumor cells showing partial or complete membrane staining at any intensity.

PD-L1 (SP142): TC scoring was based on the presence of discernible PD-L1 membrane staining of any intensity in $\geq 50\%$ of viable tumor cells.

PD-L1 (SP263): Scoring was based on percentage of viable tumor cells showing partial or complete membrane staining at any intensity.

PD-L1 TUMOR PROPORTION SCORE (TPS)			
Biomarker	Result	TPS	Threshold
PD-L1 (22c3)	Positive	90%	TPS $\geq 1\%$

PD-L1 22c3: Scoring was based on the percentage of viable tumor cells showing partial or complete membrane staining. There are three categories of PD-L1 expression defined by the PD-L1 22c3 IHC pharmDx NSCLC interpretation guide: TPS < 1% (negative), TPS $\geq 1\%$ and TPS $\geq 50\%$.

Her-2 IHC: Biopsy	
Final Score	Threshold for Positive Result
Negative 0	Intensity $\geq 3+$ and ≥ 5 tumor cells staining

PD-L1 IMMUNE CELL (IC) SCORE			
Biomarker	Result	IC	Threshold
PD-L1 (SP142)	Negative	0%	$\geq 10\%$

PD-L1 (SP142): IC scoring was based on discernible PD-L1 staining of any intensity in tumor-infiltrating immune cells covering $\geq 10\%$ of tumor area occupied by tumor cells, associated intratumoral or contiguous peritumoral stroma.

Clones used: ERBB2 (Her2/Neu) (4B5), PD-L1 (22c3), PD-L1 (28-8), PD-L1 (SP142), PD-L1 (SP263).



Electronic Signature

Jose de los Santos, MD

08/09/2024

Additional IHC results continued on the next page. >

PATIENT: [REDACTED]

TN24-206675

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Protein Expression by Immunohistochemistry (IHC)

IHC Methods

The Laboratory Developed Tests (LDT) immunohistochemistry (IHC) assays were developed and their performance characteristics determined by Caris Life Sciences. These tests have not been cleared or approved by the US Food and Drug Administration. The FDA has determined that such clearance or approval is not currently necessary. Interpretations of all immunohistochemistry (IHC) assays were performed manually by a board certified pathologist using a microscope and/or digital whole slide image(s).

The following IHC assays were performed using FDA-approved companion diagnostic or FDA-cleared tests consistent with the manufacturer's instructions: ALK (VENTANA ALK (D5F3) CDx Assay, Ventana), ER (CONFIRM anti-Estrogen Receptor (ER) (SP1), Ventana), FOLR1 (VENTANA FOLR1-2.1 RxDx, Ventana), CLDN18 (43-14A, Ventana), PR (CONFIRM anti-Progesterone Receptor (PR) (1E2), Ventana), HER2/neu (PATHWAY anti-HER-2/neu (4B5), Ventana), Ki-67 (MIB-1 pharmaDx, Dako), PD-L1 22c3 (pharmDx, Dako), PD-L1 SP142 (VENTANA, non-small cell lung cancer), PD-L1 28-8 (pharmDx, Dako, gastric / GEJ, non-small cell lung cancer), PD-L1 SP263 (Ventana, non-small cell lung cancer), and Mismatch Repair (MMR) proteins (MLH1, MSH2, MSH6, and PMS2; VENTANA MMR RxDx Panel, Ventana).

HER2 results and interpretation follow the ASCO/CAP scoring criteria. Bartley, A.N., J.A. Ajani, et al. (2016). "HER2 testing and clinical decision making in gastroesophageal adenocarcinoma: guideline from the College of American Pathologists, American Society for Clinical Pathology, and the American Society of Clinical Oncology". J Clin Oncol. 35(4):446-464.

Professional Component Performed: - - Location 1, Chandler, AZ. Jose de los Santos, MD, Lab Director.

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#	Drug	Biomarker	Reference
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2	alectinib	ALK	Gadgeel, S., D.R. Camidge, et al. (2018). "Alectinib versus crizotinib in treatment-naïve anaplastic lymphoma kinase-positive (ALK+) non-small-cell lung cancer: CNS efficacy results from the ALEX study." Ann Oncol 29 (11): 2214-2222. View Citation Online
3	alectinib, brigatinib, ceritinib, crizotinib, lorlatinib	ALK	National Comprehensive Cancer Network. NCCN Clinical Practice Guidelines in Oncology. Non-Small Cell Lung Cancer Version 1.2020
4	brigatinib	ALK	Camidge, D.R., S. Popat, et al. (2018). "Brigatinib versus Crizotinib in ALK-Positive Non-Small-Cell Lung Cancer." N Engl J Med 379(21): 2027-2039. View Citation Online
5	brigatinib	ALK	Lindeman, N.I., Y. Yatabe, et al. (2018). "Updated Molecular Testing Guideline for the Selection of Lung Cancer Patients for Treatment With Targeted Tyrosine Kinase Inhibitors Guideline From the College of American Pathologists, the International Association for the Study of Lung Cancer, and the Association for Molecular Pathology." J Thorac Oncol 13(3): 323-358. View Citation Online
6	brigatinib	ALK	Reckamp, K., J. Lee, et al. (2019). "Comparative efficacy of brigatinib versus ceritinib and alectinib in patients with crizotinib-refractory anaplastic lymphoma kinase-positive non-small cell lung cancer." Curr Med Res Opin 35(4):569-576. View Citation Online
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11	dabrafenib, dabrafenib + trametinib, encorafenib + binimetinib, vemurafenib	BRAF	Hyman, D.H., J. Baselga, et al. (2015). "Vemurafenib in Multiple Nonmelanoma Cancers with BRAF V600 Mutations." NEJM 373(8):726-736. View Citation Online
12	dabrafenib, dabrafenib + trametinib, encorafenib + binimetinib, vemurafenib	BRAF	Planchard, D., B.E. Johnson, et al. (2016). "An open-label phase II trial of dabrafenib (D) in combination with trametinib (T) in patients (pts) with previously treated BRAF V600E-mutant advanced non-small cell lung cancer (NSCLC; BRF113928)." J Clin Oncol 34: 15_suppl, 107-107. View Citation Online
13	dabrafenib, dabrafenib + trametinib, encorafenib + binimetinib, vemurafenib	BRAF	Planchard, D., B.E. Johnson, et al. (2017). "Dabrafenib plus trametinib in patients with previously untreated BRAFV600E-mutant metastatic non-small-cell lung cancer: an open-label, phase 2 trial." Lancet Oncol 18(1):1307-1316. View Citation Online
14	dabrafenib, dabrafenib + trametinib, encorafenib + binimetinib, vemurafenib	BRAF	Riely, G. J., B. E. Johnson, et al. (2023). "Phase II, Open-Label Study of Encorafenib Plus Binimetinib in Patients With BRAFV600E-Mutant Metastatic Non-Small-Cell Lung Cancer." J Clin Oncol. 41 (21): 3700-3711 View Citation Online
15	dabrafenib, dabrafenib + trametinib, encorafenib + binimetinib, vemurafenib	BRAF	Subbiah, V., D.M. Hyman, et al. (2019). "Efficacy of Vemurafenib in Patients With Non-Small-Cell Lung Cancer With BRAF V600 Mutation: An Open-Label, Single-Arm Cohort of the Histology-Independent VE-BASKET Study." JCO Precis Oncol 2019: 3, 1-9. View Citation Online

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#	Drug	Biomarker	Reference
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17	afatinib, dacomitinib	EGFR	Wu, Y-L, T.S. Mok, et al (2017). "Dacomitinib versus gefitinib as first-line treatment for patients with EGFR-mutation-positive non-small-cell lung cancer (ARCHER 1050): a randomized, open-label, phase III trial." Lancet Oncol. 2017;18(11):1454-1466. View Citation Online
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19	afatinib, dacomitinib	EGFR	Yang, J.C., et al. (2015). "Clinical activity of afatinib in patients with advanced non-small-cell lung cancer harbouring uncommon EGFR mutations: a combined post-hoc analysis of LUX-Lung 2, LUX-Lung 3, and LUX-Lung 6." Lancet Oncol; (7):830-838. View Citation Online
20	erlotinib, gefitinib	EGFR	Brugger, W., F. Cappuzzo, et. al. (2011). "Prospective molecular marker analyses of EGFR and KRAS from a randomized, placebo-controlled study of erlotinib maintenance therapy in advanced non-small-cell lung cancer." J. Clin. Oncol. 29:4113-4120. View Citation Online
21	erlotinib, gefitinib	EGFR	Fukuoka, M., T.S.K. Mok, et. al. (2011). "Biomarker analyses and final overall survival results from a phase III, randomized, open-label, first-line study of gefitinib versus carboplatin/paclitaxel in clinically selected patients with advanced non-small-cell lung cancer in Asia (IPASS). J. Clin. Oncol. DOI: 10.1200/JCO.2010.33.4235. View Citation Online
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26	adagrasib, sotorasib	KRAS	Janne, P. A., A. I. Spira, et al. (2022). "Adagrasib in Non-Small-Cell Lung Cancer Harboring a KRASG12C Mutation." N Engl J Med. 14;387(2): 120-131 View Citation Online
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PHYSICIAN: ASHISH MASKEY

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38	nivolumab/ipilimumab combination	PD-L1 (28-8)	Hellmann, M.D., S.S. Ramalingam, et al., (2019). "Nivolumab plus Ipilimumab in Advanced Non-Small-Cell Lung Cancer." <i>N Engl J Med</i> 381:2020-31. View Citation Online
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42	pralsetinib, selpercatinib	RET	National Comprehensive Cancer Network. NCCN Clinical Practice Guidelines in Oncology. Non-Small Cell Lung Cancer Version 6.2020 View Citation Online
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#	Drug	Biomarker	Reference
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